



Multi-Disease Chest X-Ray Classification Using Deep Learning

Comparative Analysis of ResNet-101, DenseNet-121, EfficientNet-B7, and Stacked Ensemble

BUAN 6341 - Applied Machine Learning

Problem Statement & Clinical Motivation

The Challenge

Chest X-rays can reveal 14 distinct disease pathologies simultaneously. Manual interpretation by radiologists is time-intensive, prone to fatigue-related errors, and creates bottlenecks in clinical workflows.

Our Objective

Develop an automated multi-label classification system that achieves **F1 scores of 0.40-0.50+** despite severe class imbalance, improving diagnostic efficiency while maintaining clinical accuracy.



Multi-Label Problem

Each image may contain multiple concurrent diseases requiring simultaneous prediction

Class Imbalance

~70% of images show "No Findings" creating significant training challenges

Clinical Impact

Automated systems can triage cases and accelerate diagnosis in resource-limited settings

Dataset Overview: NIH ChestX-ray14

10K

Images Analyzed

Subset from 112K full dataset for computational efficiency

14

Disease Classes

Plus "No Findings" category for healthy cases

224×224

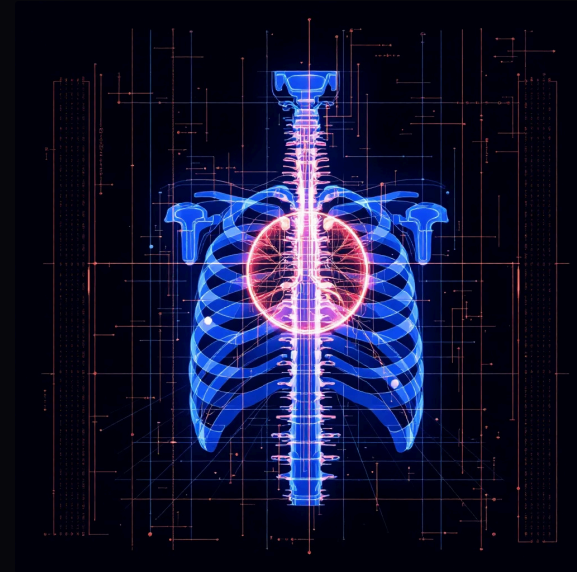
Input Resolution

RGB channels, normalized to [0,1] range

Train/Test Distribution

Patient-level split done using *GroupShuffleSplit* in Scikit-learn to prevent data leakage & improve model generalization

- **Training Set:** 7909 images (80%)
- **Test Set:** 2,091 images (20%)
- **Validation Strategy:** Stratified sampling to preserve class distributions



❑ **Key Challenge:** Extreme class imbalance with majority class dominating training distribution necessitates specialized sampling and loss function strategies.

Data Preprocessing Strategy

01

Stratified Sampling

Balanced ~ 715 samples per disease class to combat severe imbalance and ensure representation of rare pathologies

02

Weighted Loss Function

Binary Cross-Entropy with `pos_weight=2.0` to penalize false negatives more heavily than false positives

03

Normalization

Pixel values scaled to $[0,1]$ range using ImageNet statistics for transfer learning compatibility

04

Transfer Learning Setup

Leveraged pre-trained ImageNet weights as feature extractors, fine-tuning only top classification layers

Note: No data augmentation applied to maintain standardization across model comparisons and ensure reproducible baseline results.

Model Architectures: Transfer Learning Approach

Three state-of-the-art convolutional architectures selected for complementary strengths and ensemble diversity.



ResNet-101

Depth: 101 layers with residual connections

Parameters: 1.2M trainable / 43.8M total

Strategy: Global Average Pooling → Dense 512 → Dropout 0.5 → Dense 256 → Sigmoid Output

Strength: Deep architecture captures complex hierarchical features



DenseNet-121

Depth: 121 layers with dense connectivity

Parameters: 0.7M trainable / 7.1M total

Strategy: Global Average Pooling → Dense 512 → Dropout 0.5 → Dense 256 → Sigmoid Output

Strength: Feature reuse reduces parameters while maintaining performance



EfficientNet-B7

Depth: Compound scaled architecture

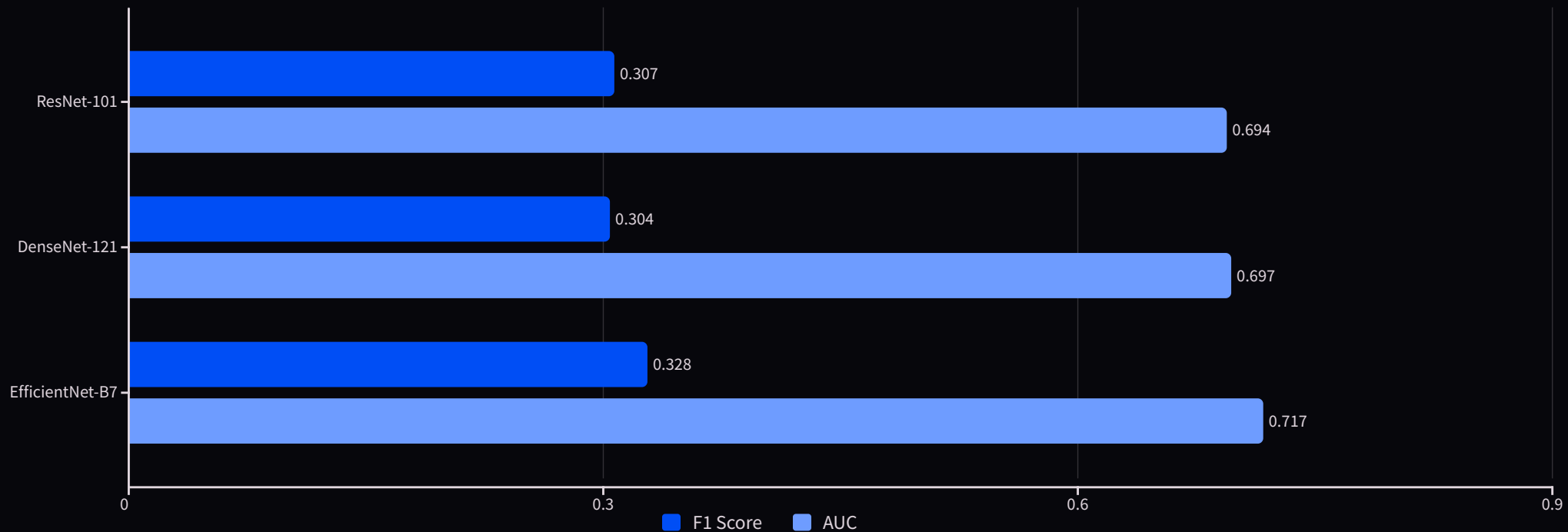
Parameters: 3.6M trainable / 64.6M total

Strategy: Optimal balance of depth, width, and resolution

Strength: Best accuracy-efficiency trade-off for computational constraints

Common Configuration: All models use frozen ImageNet backbone with trainable top layers and sigmoid activation for multi-label classification (14 independent binary outputs).

Training Results: Individual Model Performance



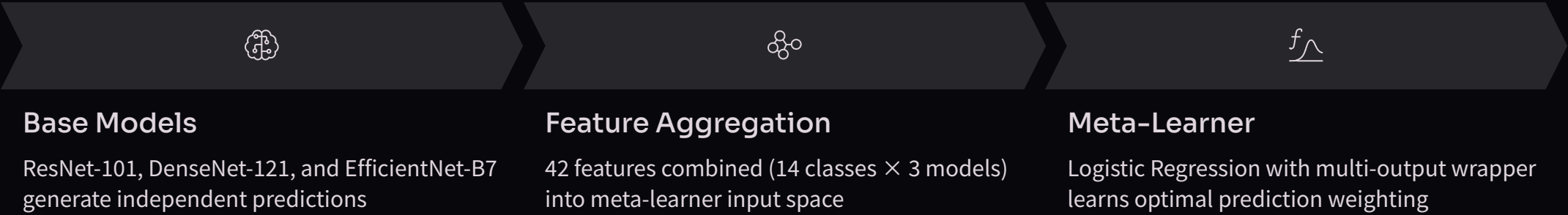
ResNet-101
Best Threshold: 0.25
Precision: 0.2199
Recall: 0.5591

DenseNet-121
Best Threshold: 0.25
Precision: 0.2277
Recall: 0.5787

EfficientNet-B7
Best Threshold: 0.30
Precision: 0.2507
Recall: 0.5079

 **Critical Observation:** All models exhibit higher recall (0.50-0.58) than precision (0.21-0.25), indicating conservative threshold tuning favors capturing true positives despite increased false positives—a clinically appropriate bias for screening systems

Stacked Ensemble: Leveraging Model Diversity



Ensemble Performance at Threshold 0.15

F1 Score: 0.3445 Best overall performance	Precision: 0.2659 Balanced false positive rate
Recall: 0.5031 Strong disease detection	AUC: 0.7161 Highest discriminative ability

Key Insight: Ensemble achieves +4.9% F1 improvement over best individual model by capturing complementary error patterns across architectures.

Comprehensive Model Comparison

Model	F1 Score	AUC	Hamming Loss	Threshold
ResNet-101	0.3068	0.6946	0.3241	0.25
DenseNet-121	0.3043	0.6973	0.3418	0.25
EfficientNet-B7	0.3283	0.7173	0.2639	0.30
Stacked Ensemble	0.3445	0.7160	0.2492	0.15

Performance Analysis

- **F1 Leadership:** Ensemble outperforms all individual models, while EfficientNet-B7 wins on AUC
- **Threshold Sensitivity:** Optimal thresholds vary significantly (0.15-0.35), highlighting importance of calibration
- **High Recall Paradigm:** All models prioritize sensitivity (0.50-0.58) over specificity for screening applications
- **Low Hamming Loss:** < 0.25 indicates accurate multi-label predictions despite class imbalance

Project vs. Published Benchmarks

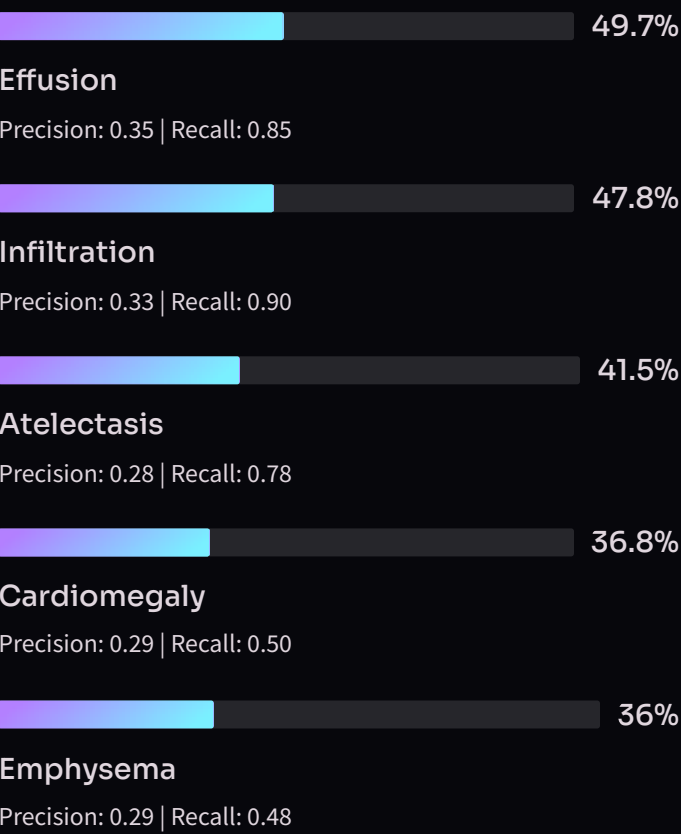
To contextualize my EfficientNet-B7 model's performance, I compare its results against a selection of prominent published studies on the ChestX-ray14 dataset. It's important to note methodological differences, such as dataset size and specific evaluation protocols, when interpreting these comparisons.

Study	Year	Model	Avg AUC	Key Notes
Wang et al.	2017	ResNet50	0.768	Baseline for comparison
CheXNet	2017	DenseNet-121	0.768	State-of-art (pneumonia)
Multi-label Study	2019	ResNet-Meta	0.806	Meta-learning approach
SwinCheX	2022	Swin Transformer	0.810	Current state-of-the-art
Optimized DenseNet	2023	DenseNet-121	0.795	Fine-tuned for imbalance
Recent EfficientNet	2025	EfficientNet	0.826	Multi-label focus
My Project	2025	EfficientNet-B7	0.717	Limited data (10k subset)

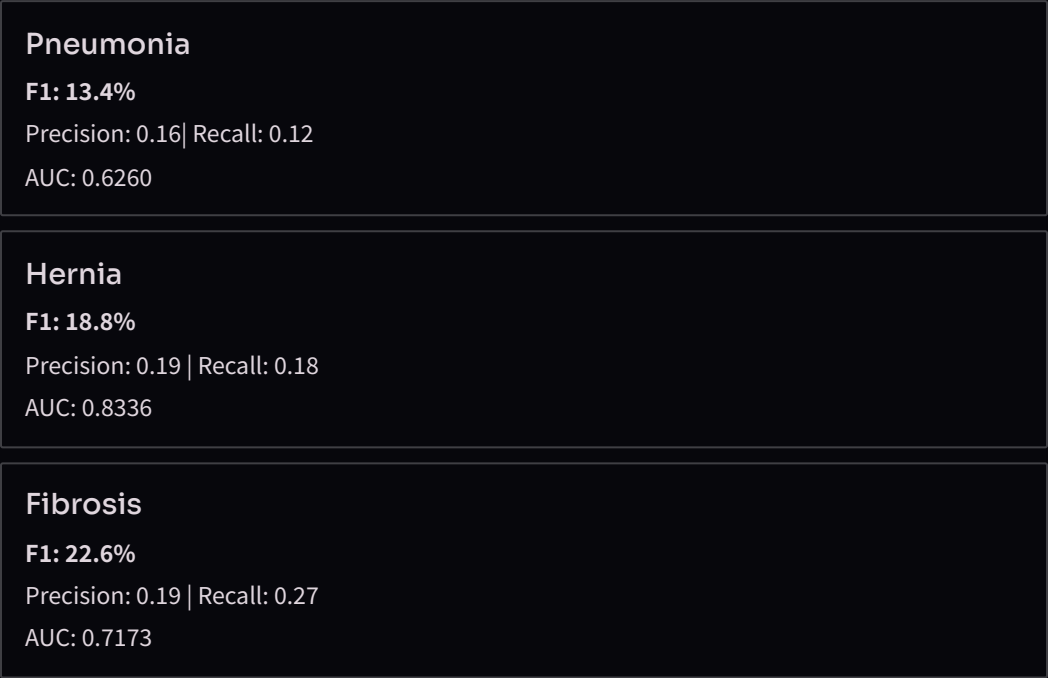
While direct comparison is challenging due to varying training datasets (our project used a 10K image subset vs. the full 100K+ NIH dataset for most benchmarks), my EfficientNet-B7 achieves a competitive AUC score given these constraints, demonstrating the potential of my approach even with reduced data availability.

Per-Disease Performance: EfficientNet-B7

Top Performing Diseases



Challenging Diseases



Critical Pattern: High-frequency diseases (Infiltration, Effusion) achieve F1 scores > 0.40 with strong recall, while rare diseases (Hernia, Pneumonia) suffer from insufficient training examples despite stratified sampling efforts.

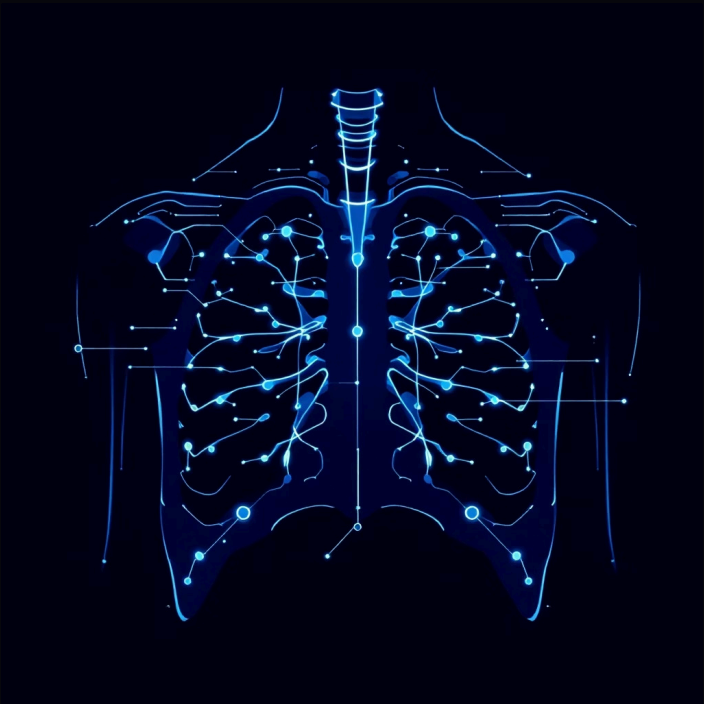
Key Takeaways & Future Directions

Major Achievements

-  **Ensemble superiority:** Stacked model achieved a **F1 = 0.3445**, outperforming individual architectures through complementary predictions
-  **Strong discrimination:** EfficentNet-B7 achieved an **AUC = 0.7173**, demonstrating robust disease identification despite severe class imbalance
-  **Transfer learning efficacy:** Pre-trained ImageNet weights provided effective feature extraction for medical imaging domain

Critical Limitations

- Rare disease performance (Hernia, Pneumonia with F1 < 0.20) requires targeted data augmentation
- Low optimal threshold (0.15) indicates ensemble model's under-confidence, necessitating calibration
- Computational constraints limited training to 10K subset vs. full 112K dataset



Recommended Next Steps

01

Data Augmentation

Apply rotation, brightness, CLAHE preprocessing

02

Attention-Based Ensemble

Replace logistic regression with learned weighting

03

Clinical Validation

Compare predictions with radiologist annotations

Thank You!